PROJECT SPECIFICATION

Subgoal Planning under Partial Observability for Legged Navigation over Sparse Terrain

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Abstract

Legged robots promise access to terrain that wheeled platforms cannot traverse, yet sparse-foothold locomotion, i.e. crossing stepping stones, separated platforms or narrow piles, remains one of the most demanding problems in robot learning [1]. Recent work shows that a single learned policy can cross such terrain when locomotion is posed as reaching a goal position rather than tracking a velocity [1, 2]. There, however, goals lie a few metres away and every direction is walkable. Real tasks break both assumptions: the robot perceives only its surroundings, and the direct line to a distant goal may end in a gap or a dead end that it cannot yet see. Choosing where to go next then becomes a planning problem under *partial observability*. Layered systems, in which a navigation module issues targets to learned low-level skills [3], suggest a natural decomposition. This project investigates it at student scale: a simulated quadruped must reach a fixed goal 6–10 m away across a field of piles while perceiving only about 2 m around itself.

The system comprises two layers. A perceptive goal-reaching locomotion policy, trained with proximal policy optimisation [4] in massively parallel simulation [5] and a smooth terrain curriculum [6] over gap width, pile size, height and slant, walks to subgoals 1–2 m away using proprioception and a ray-cast height map. Above it, a subgoal planner picks the next pile from a 2 m local window whenever a subgoal is reached or times out. Four planners are compared: a memoryless greedy planner; an optimistic replanner that runs A* [7] over all piles seen so far under the free-space assumption [8]; a learned planner that scores the visible piles, trained by imitating a privileged planner with DAgger [9, 10] and optionally fine-tuned with reinforcement learning; and the privileged planner itself, A* on the full map, used only as teacher and upper bound. Because a local student cannot see what its full-map teacher sees [11], the learned planner is tested with and without memory (recurrent state, visit counts) to prevent looping in dead ends. Planners are developed in an abstract pile-graph simulator calibrated with the controller's measured hop success, and evaluated end to end in MuJoCo [12] through the GPU-accelerated mjlab framework [13] for a Unitree Go1 model.

The study addresses three research questions. (i) Does a smooth, performance-adaptive curriculum yield a more capable controller than uniform sampling of terrain difficulty under an identical budget? (ii) What is the *price of partial observability*: how far do local planners fall below the full-map oracle as dead ends grow deeper than the perception range? (iii) Can a planner learned by imitating a privileged oracle match the classical replanner, and which form of memory stops it from looping? Evaluation reports success rate, fall rate, time-to-goal, path efficiency and loop statistics over multiple random seeds on held-out layouts, both in the abstract simulator and in physics. The project is scoped for two weeks on partitioned NVIDIA H100 GPUs; the planners need no GPU training. It aims to provide an interpretable and reproducible testbed for studying how local perception, memory and privileged training interact when a learned controller is steered by a planner.

References

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1 Target Grade and Committed Deliverables

The group targets **grade A**. The grade-C foundation is an integrated goal-to-footstep pipeline (the controller driven by the greedy and optimistic planners) evaluated end to end in physics on held-out layouts with at least three random seeds. It comprises (a) a comparison of the smooth curriculum with uniform difficulty sampling under an identical budget, (b) a comparison of the greedy and optimistic planners against the full-map oracle, and (c) loop and dead-end-escape statistics for every planner. On top of this foundation, the group commits to three grade-A contributions, one per member:

A1 **Learned subgoal planner:** train a planner by DAgger imitation of the full-map oracle, with memory variants and optional RL fine-tuning, and test in physics whether it matches the optimistic replanner.

A2 **Curriculum study:** compare a smooth, performance-adaptive curriculum with uniform difficulty sampling under an identical training budget, and identify which terrain dimension limits the controller.

A3 **Price of partial observability:** measure how far local planners fall below the oracle as dead ends grow from 0.5 to 2 times the perception range, in the abstract simulator and in physics.

2 Work Decomposition

Member	Background	Primary ownership	Grade-A contribution
Amedeo Zaccone	Robotics, Systems & Control	Simulation environment in mjlab: Go1 asset, pile terrain (gap, size, height, slant), height-scan check, curriculum, goal-reaching rewards, PPO training runs, hop-success measurement.	**A2**: smooth curriculum vs. uniform difficulty sampling.
Damien Bardina	Machine Learning	Controller training setup with A.Z. (symmetry, chained targets); learned planner: pile-scoring network, DAgger from the oracle, memory variants, RL fine-tuning in the abstract simulator.	**A1**: learned subgoal planner by privileged imitation.
Tanmay Kumar	Machine Learning	Interface contract; abstract pile-graph simulator; oracle, greedy and optimistic planners; executor in mjlab; loop metrics; evaluation harness and statistics.	**A3**: price of partial observability.

3 Roadmap (10 working days)

Phase	A.Z. (Robotics & Control)	D.B. (ML)	T.K. (ML)	Milestone / exit criterion
Days 1–3	Toolchain smoke test on H100; Go1 asset; pile terrain; goal-reaching environment; first PPO runs on open piles.	Training setup with A.Z. (symmetry, chained targets); planner network and DAgger loop in the abstract simulator.	Interface contract; abstract simulator; oracle, greedy and optimistic planners; loop metrics.	**M1 (Day 3):** controller reaches 1–2 m targets on open piles; planners run in the abstract simulator.
Days 4–6	**A2**: smooth vs. uniform curriculum, 3 seeds each; level calibration; reward tuning.	**A1**: DAgger for memory variants, 3 seeds; RL fine-tuning; imitation-gap analysis.	**A3** in the abstract simulator; executor in mjlab; hop-success table.	**M2 (Day 6):** controller frozen; abstract simulator calibrated.
Days 7–8	**A2** evaluation on all levels; simulation videos.	Best planners retrained on the calibrated simulator; physics runs.	All planners on the physics navigation set; depth curves.	**M3 (Day 8):** experiments complete; feature freeze; all figures generated.
Days 9–10	Report: controller and curriculum.	Report: learned planner.	Report: planners and evaluation; rehearsal with cross-questioning (all).	**Final:** report, code and presentation submitted.

Compute allocation: controller RL runs on the partitioned H100 GPUs (ray-cast height scans, so no rendering is required). Planners train on CPU in the abstract simulator and use the GPUs only for physics evaluation. All runs are checkpointed and resumable; fallback toolchain: MuJoCo Playground (MJX/JAX).

4 Risk Assessment: Potential Project Killers

#	Risk	Likelihood / Impact	Early warning	Mitigation and fallback
1	Controller fails to hop reliably on hard piles within the compute budget, limiting the physics evaluation.	High / High	Cannot reach 1–2 m targets on open piles by M1; success plateaus on intermediate levels.	Prioritise H100 time for the controller and freeze it at M2 regardless. Fallback: wider piles without slant; planner studies continue in the abstract simulator.
2	Abstract simulator is too optimistic, so planner rankings do not transfer to physics.	Medium / High	Oracle success in physics far below its abstract success at M2.	Calibrate hop success over length, gap, height and slant; restrict subgoals to piles within 90° of the heading; report the remaining gap as a result.
3	Learned planner imitates the oracle poorly or loops in dead ends (imitation gap), weakening A1.	Medium / Medium	DAgger accuracy plateaus; repeated subgoals stay frequent for every memory variant.	Richer per-pile features; visit-count memory; RL fine-tuning. Fallback: report the imitation gap with trajectory analysis.